

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of specific leaf area (SLA) per plot (esy4 = 4qm plots)

MB

2026-05-29

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```
library(DHARMA)
library(emmeans)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.sla)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 234.5920 230.8435 227.0950
```

```
median(sites$y)
```

```
## [1] 232.03
```

```
sd(sites$y)
```

```
## [1] 32.48951
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 173.985 282.840
```

```
sites %>% distinct(id.site) %>% nrow()
```

```
## [1] 119
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      101
## 2 664       91
## 3 686       99
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord> <int>
## 1 positive      45
## 2 restored     221
## 3 negative      25
```

```
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22     210
## 2 R1A      81
```

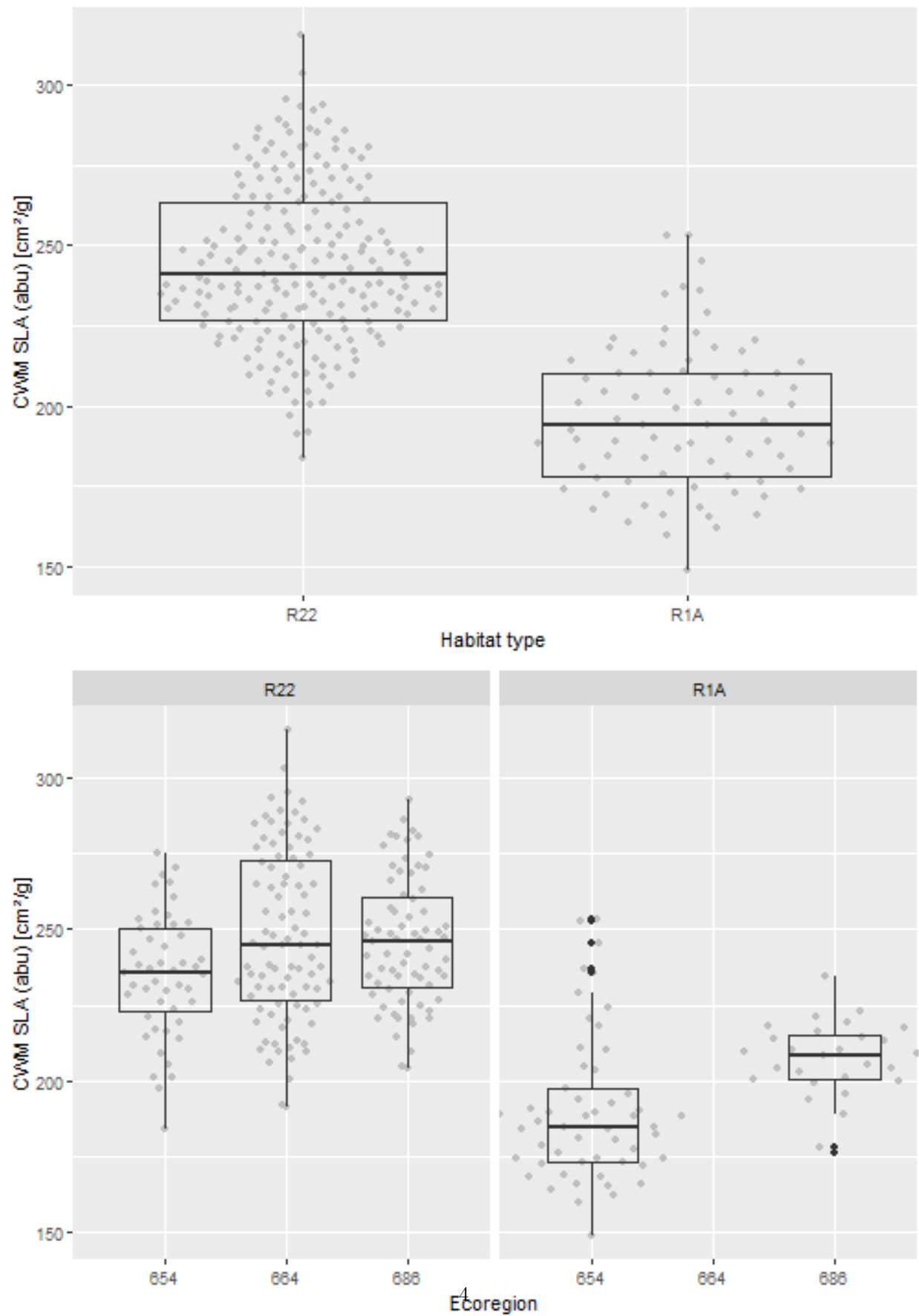
```
sites %>% count(esy4, eco.id)
```

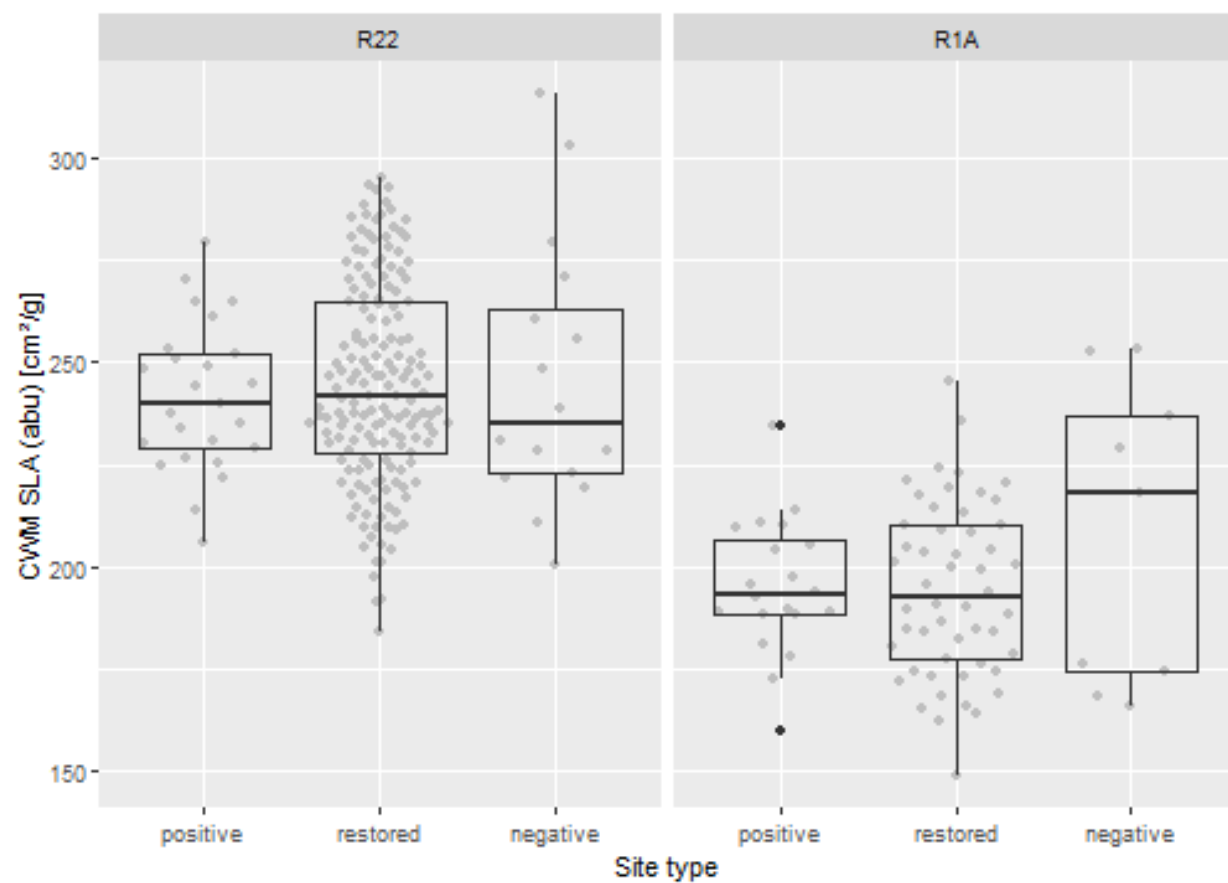
```
## # A tibble: 5 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R22  654      48
## 2 R22  664      91
## 3 R22  686      71
## 4 R1A  654      53
## 5 R1A  686      28
```

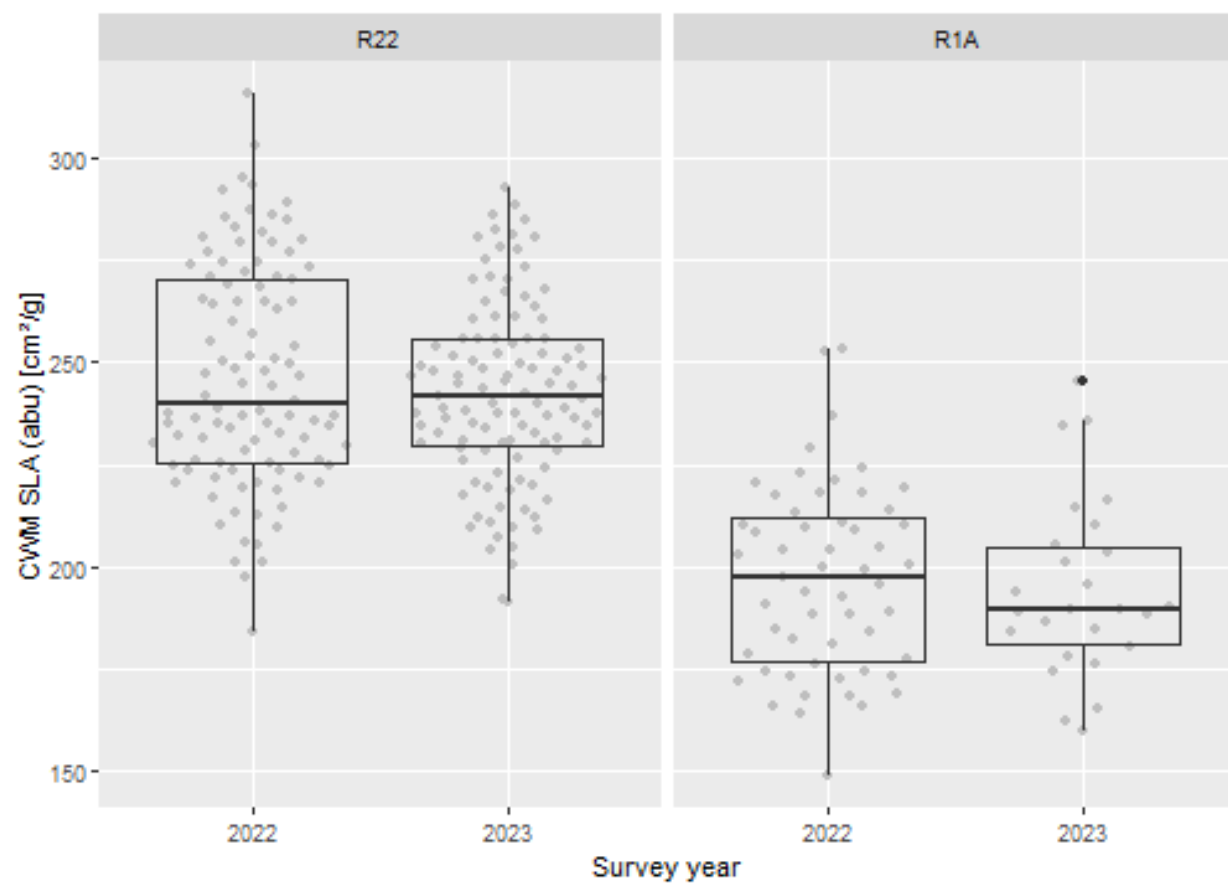
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 6 x 3
##   esy4 site.type      n
##   <fct> <ord> <int>
## 1 R22 positive      25
## 2 R22 restored     169
## 3 R22 negative      16
## 4 R1A positive      20
## 5 R1A restored      52
## 6 R1A negative       9
```

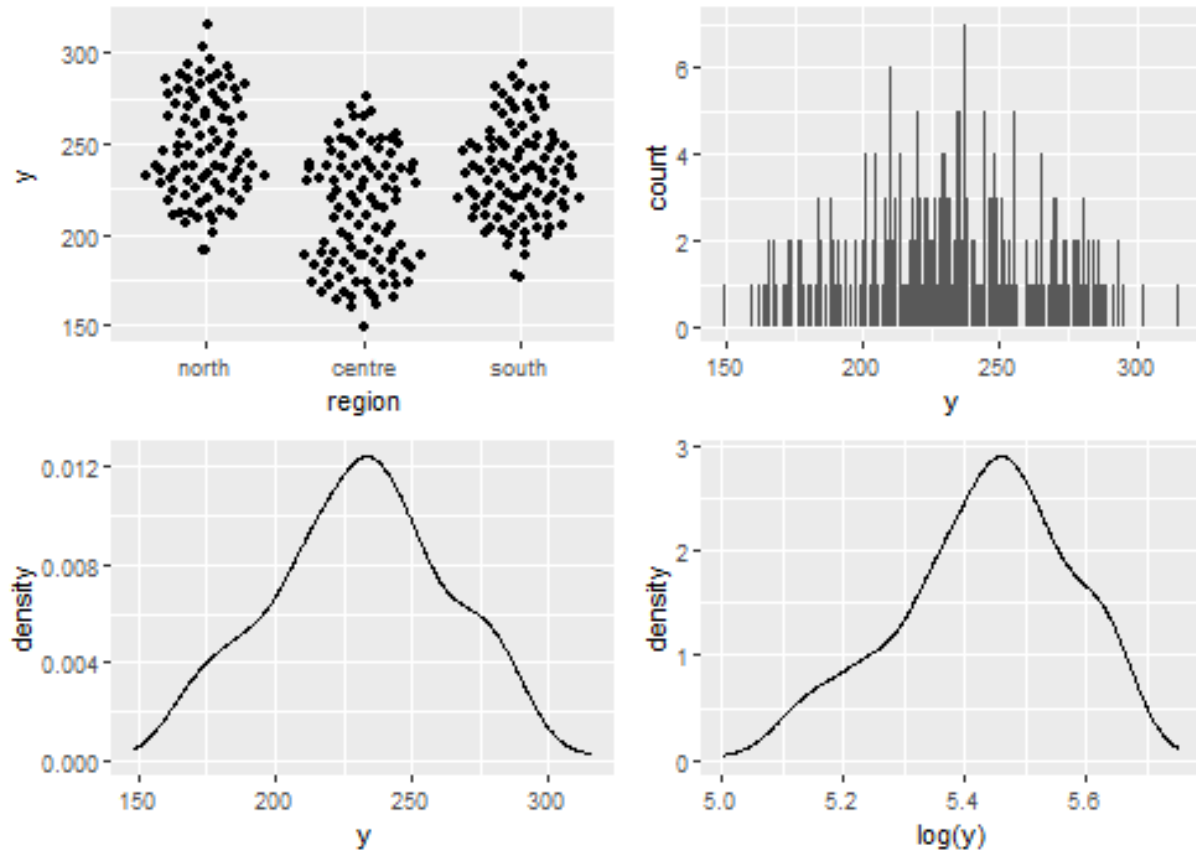
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 *Ecography* DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_sla_esy4_cwm_2.Rdata"))
load(file = here("outputs", "models", "model_sla_esy4_cwm_3.Rdata"))
m_1 <- m2
m_2 <- m3

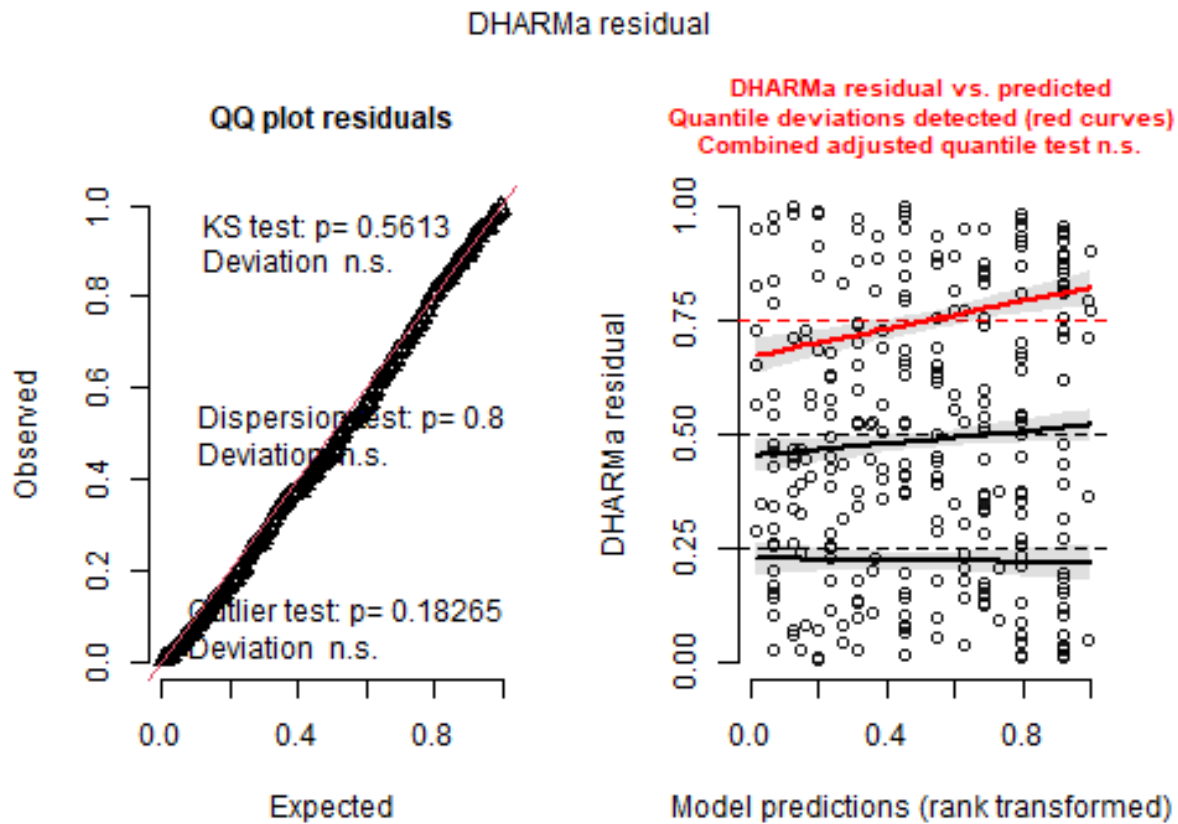
m_1@call
## lmer(formula = sqrt(y) ~ (esy4 + site.type + eco.id + obs.year)^2 +
##       (1 | id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = sqrt(y) ~ (esy4 + site.type + eco.id + obs.year +
##      hydrology)^2 + (1 | id.site), data = sites, REML = FALSE)
```

Model check

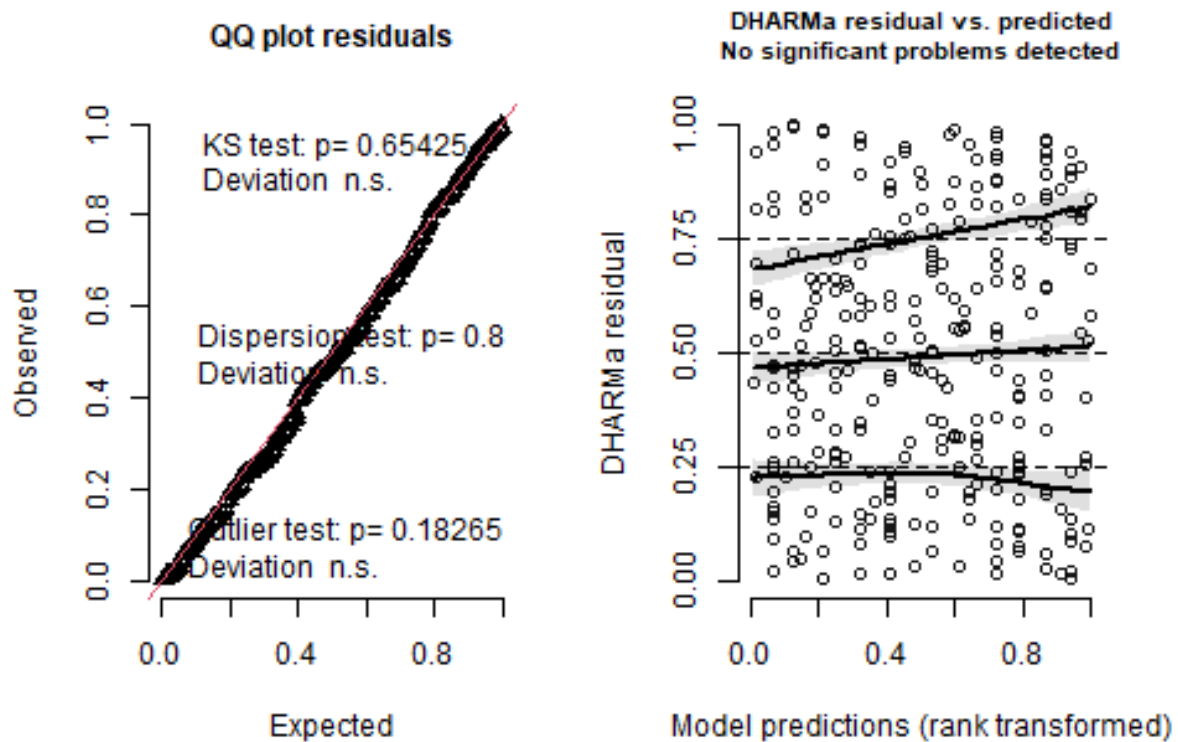
DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

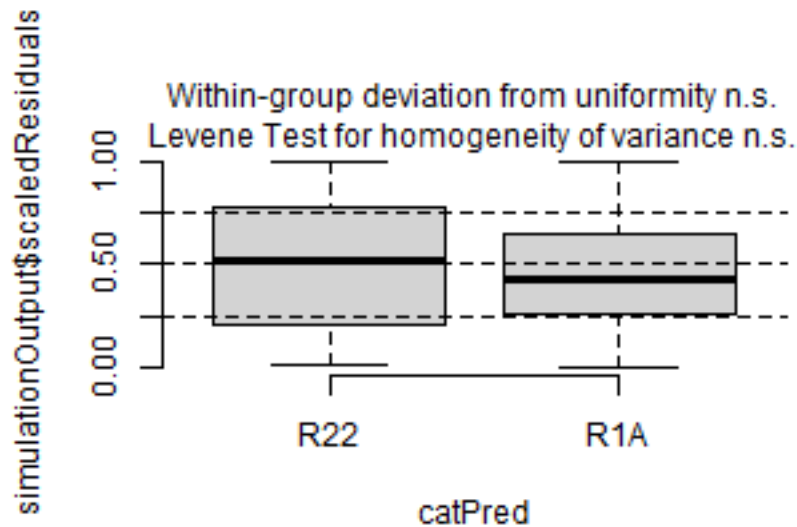


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

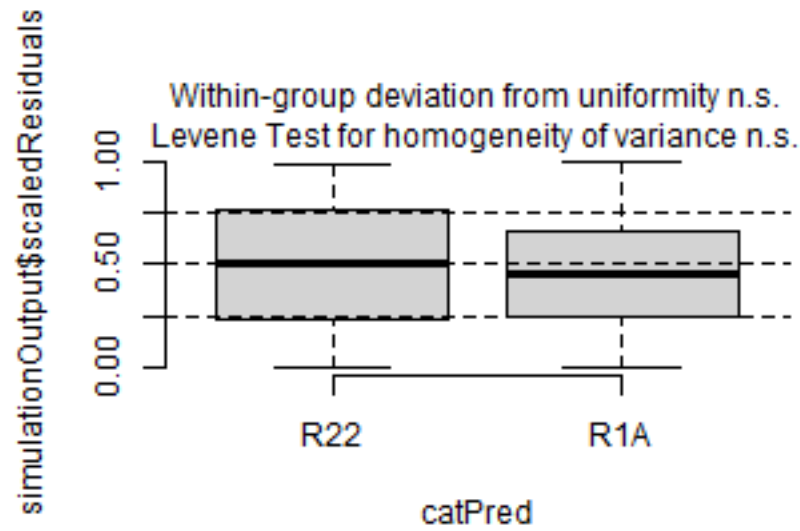
DHARMa residual



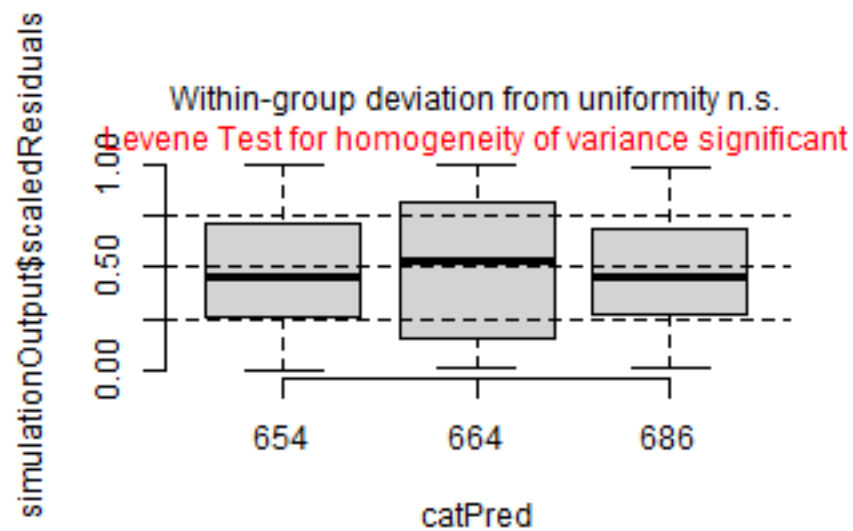
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



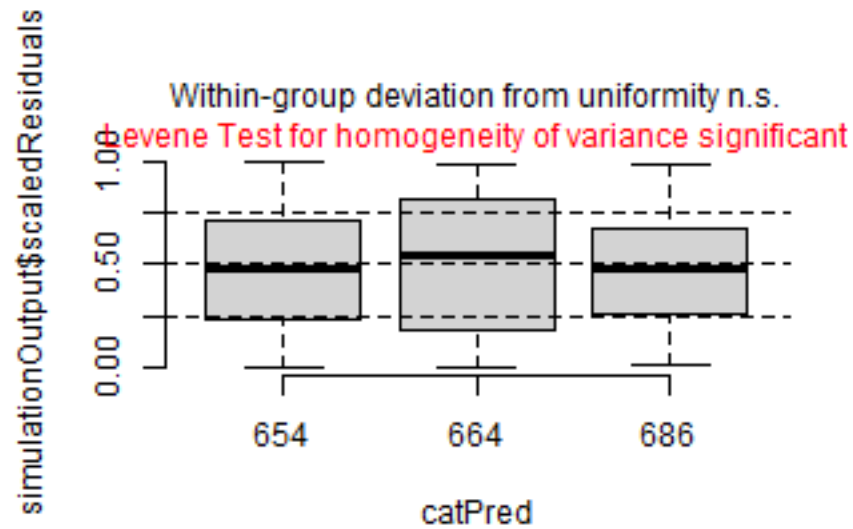
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



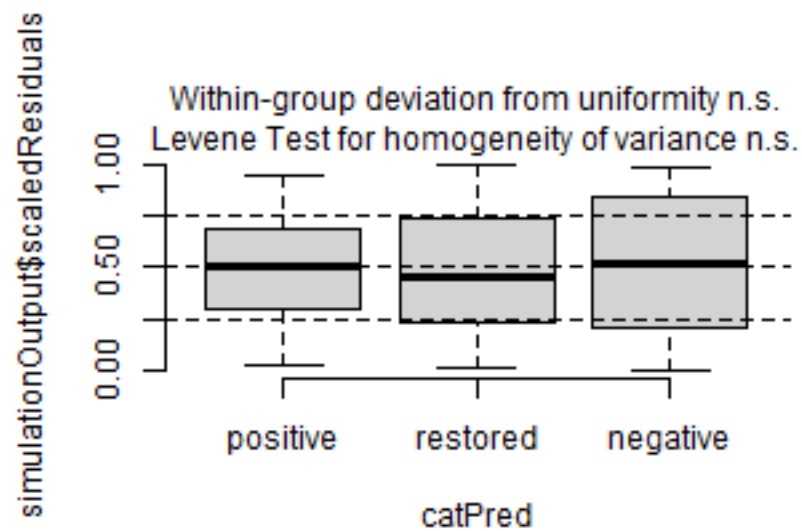
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



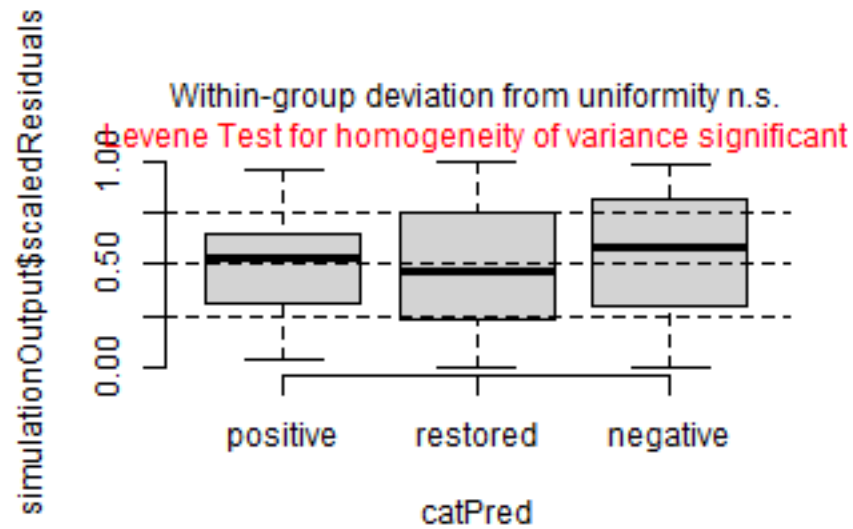
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



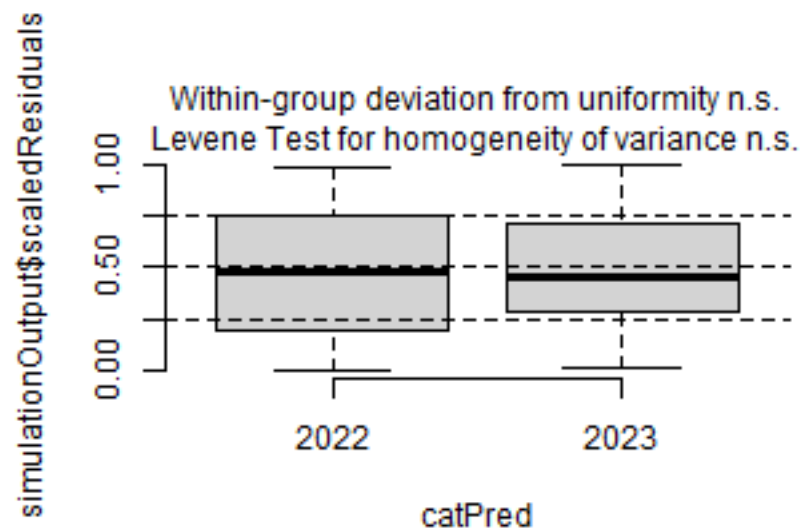
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



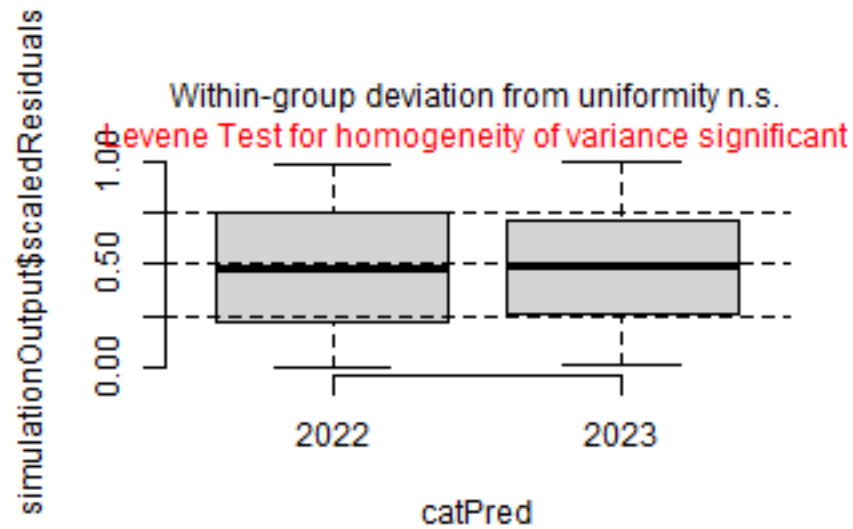
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



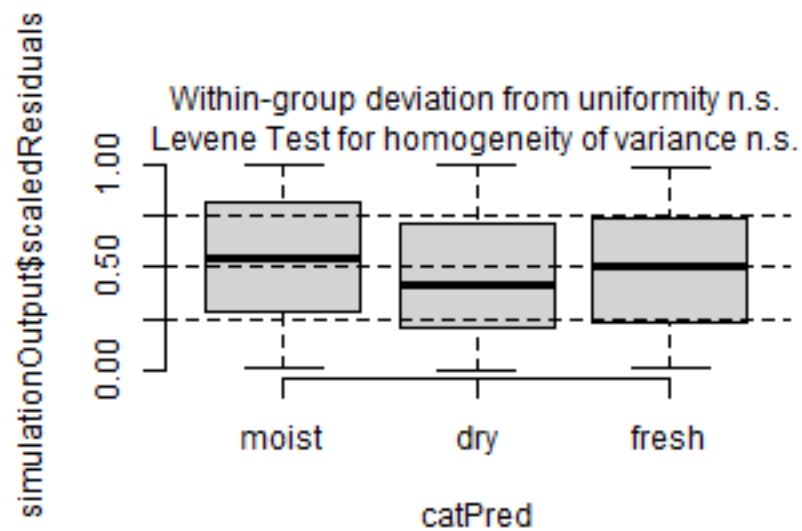
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



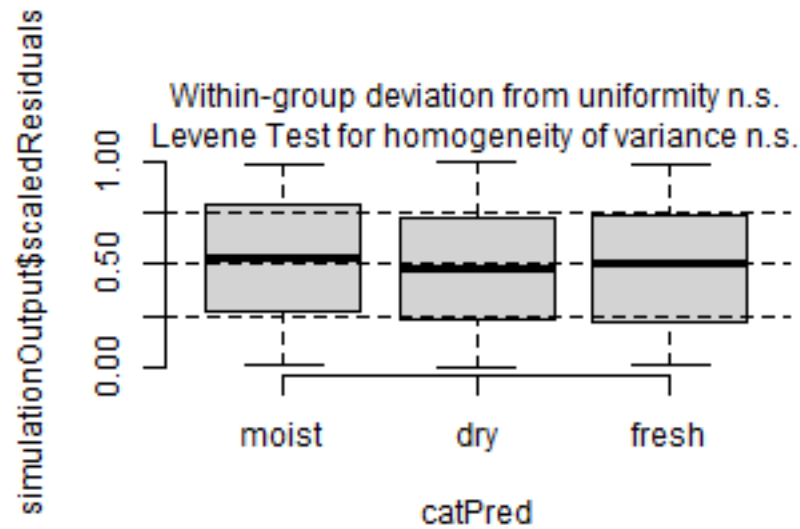
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



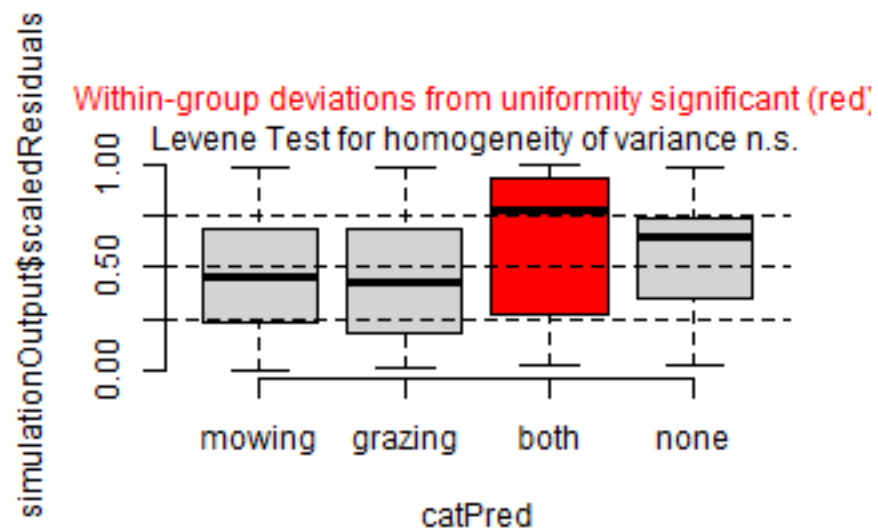
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



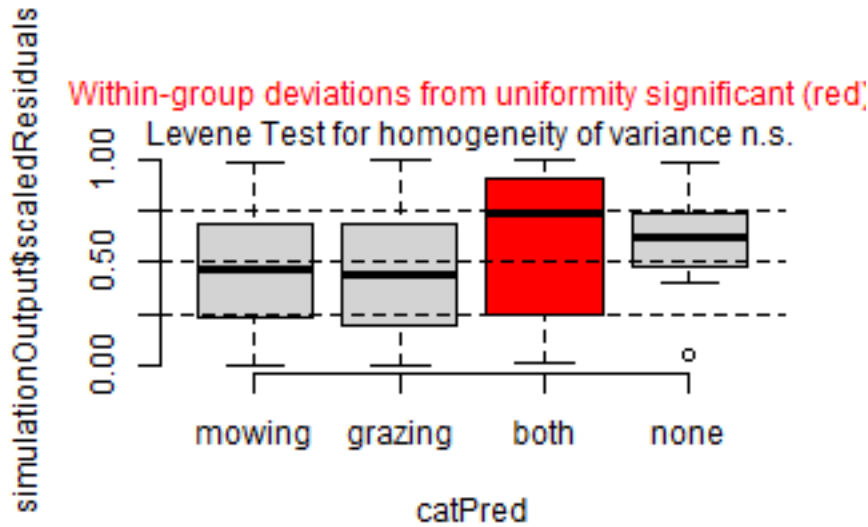
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## esy4          5.565398 1      2.359110
## site.type     59.808391 2      2.780933
## eco.id        21.621706 2      2.156366
## obs.year      6.272316 1      2.504459
## esy4:site.type 9.503280 2      1.755773
## esy4:eco.id    2.591735 1      1.609887
## esy4:obs.year  2.989081 1      1.728896
## site.type:eco.id 80.706021 4      1.731264
## site.type:obs.year 9.596825 2      1.760078
## eco.id:obs.year 11.546974 2      1.843389
```

```
car::vif(m_2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## esy4          70.853521 1      8.417453
## site.type     610.109338 2      4.969949
## eco.id        747.809303 2      5.229350
## obs.year      15.454263 1      3.931191
## hydrology     571.553093 2      4.889497
## esy4:site.type 90.941661 2      3.088095
## esy4:eco.id    22.256369 1      4.717666
## esy4:obs.year  5.791756 1      2.406607
## esy4:hydrology 11.653035 2      1.847608
## site.type:eco.id 186.649530 4      1.922553
## site.type:obs.year 15.336189 2      1.978925
## site.type:hydrology 778.808602 4      2.298417
## eco.id:obs.year 13.440637 2      1.914719
## eco.id:hydrology 2175.750423 4      2.613371
## obs.year:hydrology 29.116885 2      2.322931
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##           R2m      R2c
## [1,] 0.5488241 0.76601
MuMIn::r.squaredGLMM(m_2)
##           R2m      R2c
## [1,] 0.5910165 0.7728105
```

AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##      df      AICc
## m_1 21 616.9389
## m_2 35 634.0819
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)

## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: sqrt(y)
##              Chisq Df Pr(>Chisq)
## (Intercept)    265.9231  1 < 2.2e-16 ***
## esy4            3.5112  1  0.060954 .
## site.type       5.6904  2  0.058123 .
## eco.id          6.0946  2  0.047487 *
## obs.year        0.5316  1  0.465924
## hydrology       5.4897  2  0.064258 .
## esy4:site.type   2.5044  2  0.285876
## esy4:eco.id      1.4980  1  0.220985
## esy4:obs.year    1.7528  1  0.185521
## esy4:hydrology   0.4907  2  0.782433
## site.type:eco.id 2.7708  4  0.596887
## site.type:obs.year 11.0473 2  0.003991 **
## site.type:hydrology 3.9100 4  0.418322
## eco.id:obs.year  13.3399 2  0.001268 **
## eco.id:hydrology  5.2063 4  0.266781
## obs.year:hydrology 3.9790 2  0.136763
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_2)

## Linear mixed model fit by maximum likelihood ['lmerMod']
## Formula: sqrt(y) ~ (esy4 + site.type + eco.id + obs.year + hydrology)^2 +
##          (1 | id.site)
## Data: sites
```

```

##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    624.2    752.8   -277.1    554.2     256
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.45716 -0.58034  0.05753  0.53583  3.06183
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
## id.site   (Intercept) 0.2059   0.4537
## Residual                0.2573   0.5072
## Number of obs: 291, groups: id.site, 119
##
## Fixed effects:
##
##              Estimate Std. Error t value
## (Intercept)    15.67994   0.96154  16.307
## esy4R1A        -1.79301   0.95687  -1.874
## site.type.L      1.96898   0.82558   2.385
## site.type.Q      0.34911   0.45496   0.767
## eco.id664        0.76274   0.97739   0.780
## eco.id686       -0.34725   0.86259  -0.403
## obs.year2023     0.30103   0.41286   0.729
## hydrologyfresh  -0.38711   0.94533  -0.409
## hydrologymoist  -1.35369   0.96733  -1.399
## esy4R1A:site.type.L -1.25335   0.84786  -1.478
## esy4R1A:site.type.Q -0.08397   0.45941  -0.183
## esy4R1A:eco.id686   0.98488   0.80469   1.224
## esy4R1A:obs.year2023 -0.51759   0.39095  -1.324
## esy4R1A:hydrologyfresh 0.50583   0.84222   0.601
## esy4R1A:hydrologymoist -0.21882   0.74695  -0.293
## site.type.L:eco.id664 -0.57021   0.55724  -1.023
## site.type.Q:eco.id664  0.10617   0.34276   0.310
## site.type.L:eco.id686 -0.39565   0.46793  -0.846
## site.type.Q:eco.id686 -0.29216   0.28089  -1.040
## site.type.L:obs.year2023 -1.16428   0.39276  -2.964
## site.type.Q:obs.year2023 -0.44899   0.24435  -1.837
## site.type.L:hydrologyfresh -0.68746   0.77078  -0.892
## site.type.Q:hydrologyfresh 0.02934   0.42174   0.070
## site.type.L:hydrologymoist -0.96847   0.76780  -1.261
## site.type.Q:hydrologymoist -0.56032   0.45224  -1.239
## eco.id664:obs.year2023 -1.16883   0.32056  -3.646
## eco.id686:obs.year2023 -0.58313   0.28008  -2.082
## eco.id664:hydrologyfresh 0.33061   0.90517   0.365
## eco.id686:hydrologyfresh 0.99250   0.83285   1.192
## eco.id664:hydrologymoist 0.87051   1.00790   0.864
## eco.id686:hydrologymoist 1.13969   0.90056   1.266
## obs.year2023:hydrologyfresh -0.21757   0.34179  -0.637
## obs.year2023:hydrologymoist 0.56476   0.44367   1.273
##
## Correlation matrix not shown by default, as p = 33 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it

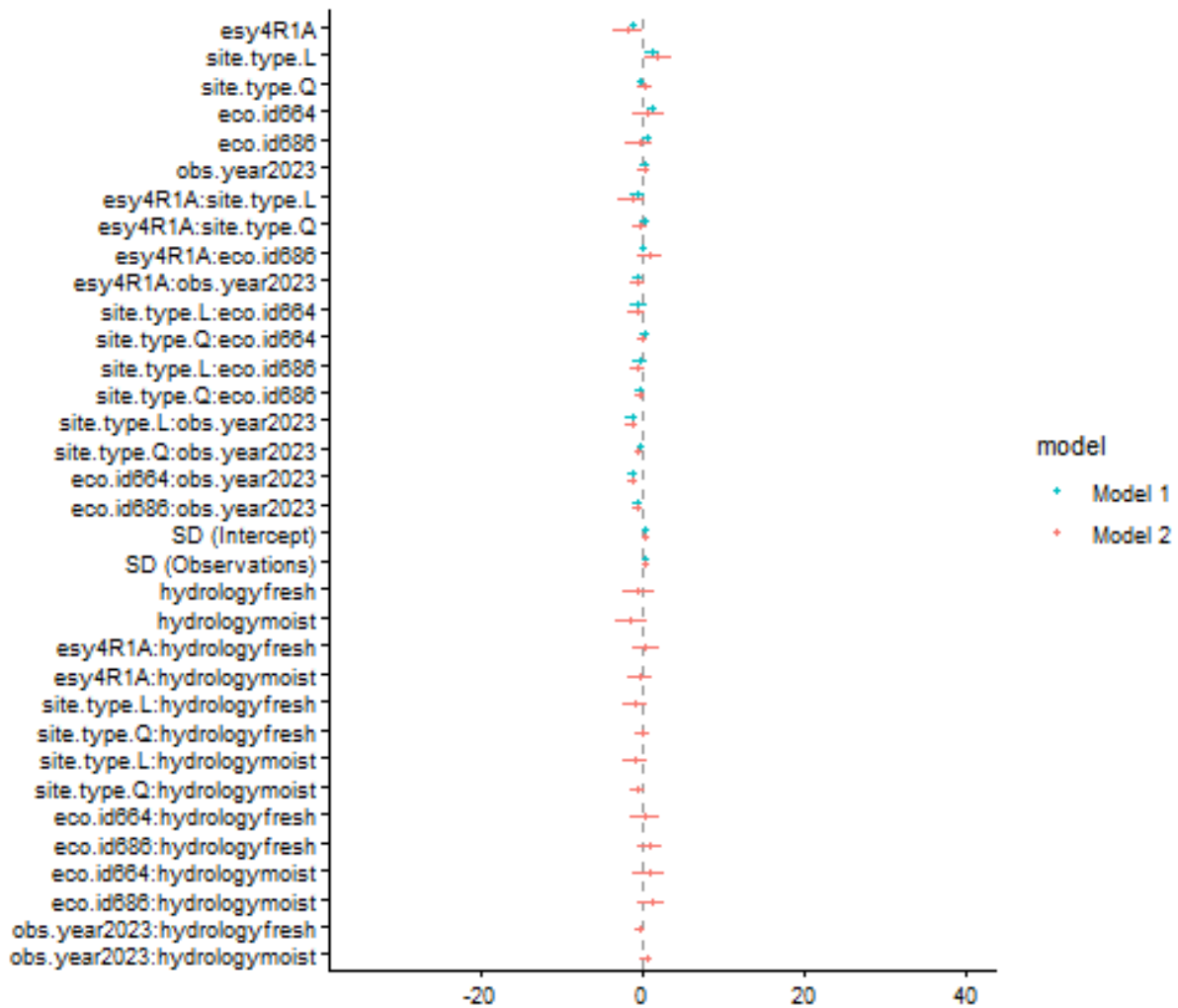
```

```
## fit warnings:  
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Forest plot

```
dotwhisker::dwplot(  
  list(m_1, m_2),  
  ci = 0.95,  
  show_intercept = FALSE,  
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +  
  xlim(-35, 40) +  
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals  
##   for random effect parameters.  
## Package 'merDeriv' needs to be installed to compute confidence intervals  
##   for random effect parameters.  
  
## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.  
## i Please use the `linewidth` aesthetic instead.  
## i The deprecated feature was likely used in the dotwhisker package.  
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.  
## This warning is displayed once every 8 hours.  
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was  
## generated.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
# (emm <- emmeans(
#   m_2,
#   reupairwise ~ esy4,
#   type = "response"
# ))
# plot(emm, comparison = TRUE)
```

ESY EUNIS Habitat type

```
# (emm <- emmeans(
#   m_2,
#   reupairwise ~ eco.id,
#   type = "response"
```

```
# ))
# plot(emm, comparison = TRUE)
```

Habiat type x Region

```
# (emm <- emmeans(
#   m_2,
#   revpairwise ~ site.type,
#   type = "response"
# ))
# plot(emm, comparison = TRUE)
```

Habiat type x Site type

Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMA_0.4.7    patchwork_1.3.2  ggbeeswarm_0.7.2
## [5] lubridate_1.9.4  forcats_1.0.0    stringr_1.5.2    dplyr_1.1.4
## [9] purrr_1.1.0      readr_2.1.5      tidyr_1.3.1      tibble_3.3.0
## [13] ggplot2_4.0.0    tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6        magrittr_2.0.4      multcomp_1.4-28
## [7] compiler_4.5.2     mgcv_1.9-3          vctrs_0.6.5
## [10] pkgconfig_2.0.3    crayon_1.5.3        fastmap_1.2.0
## [13] backports_1.5.0    labeling_0.4.3      utf8_1.2.6
## [16] ggstance_0.3.7     promises_1.3.2      rmarkdown_2.30
## [19] tzdb_0.5.0         nloptr_2.2.1        bit_4.6.0
## [22] xfun_0.53          later_1.4.2         parallel_4.5.2
## [25] R6_2.6.1           gap.datasets_0.0.6  stringi_1.8.7
## [28] qgam_2.0.0         RColorBrewer_1.1-3  car_3.1-3
## [31] boot_1.3-32        estimability_1.5.1  Rcpp_1.1.0
```

## [34] iterators_1.0.14	knitr_1.50	zoo_1.8-14
## [37] parameters_0.25.0	httpuv_1.6.16	Matrix_1.7-4
## [40] splines_4.5.2	timechange_0.3.0	tidyselect_1.2.1
## [43] rstudioapi_0.17.1	abind_1.4-8	yaml_2.3.10
## [46] MuMIn_1.48.11	doParallel_1.0.17	codetools_0.2-20
## [49] lattice_0.22-7	plyr_1.8.9	bayestestR_0.15.3
## [52] shiny_1.10.0	withr_3.0.2	S7_0.2.0
## [55] evaluate_1.0.5	marginaleffects_0.25.1	survival_3.8-3
## [58] pillar_1.11.1	gap_1.6	carData_3.0-5
## [61] foreach_1.5.2	stats4_4.5.2	reformulas_0.4.1
## [64] insight_1.2.0	generics_0.1.4	vroom_1.6.6
## [67] rprojroot_2.0.4	hms_1.1.3	scales_1.4.0
## [70] minqa_1.2.8	xtable_1.8-4	glue_1.8.0
## [73] tools_4.5.2	data.table_1.17.8	lme4_1.1-37
## [76] mvtnorm_1.3-3	grid_4.5.2	rbibutils_2.3
## [79] datawizard_1.1.0	nlme_3.1-168	Rmisc_1.5.1
## [82] performance_0.13.0	beeswarm_0.4.0	vipor_0.4.7
## [85] Formula_1.2-5	cli_3.6.5	gtable_0.3.6
## [88] digest_0.6.37	TH.data_1.1-3	farver_2.1.2
## [91] htmltools_0.5.8.1	lifecycle_1.0.4	mime_0.13
## [94] bit64_4.6.0-1	dotwhisker_0.8.4	MASS_7.3-65